

Thailand Semiconductor Supply Chain Risk Assessment

Assessing Thailand's Exposure to Taiwan Semiconductor Supply Chain Disruption

One-Page Project Summary | Methakan Bualuang | Industrial Engineering, Chulalongkorn University | June 2026

43.6% Taiwan proxy dependency in 2024 Up from 22.8% in 2014	USD 8.35B Base-case semiconductor-linked exposure Across four Thai industries	USD 5.77B Electronics exposure ~69% of total base-case RAR	USD 3.34B Severe scenario impact 40% disruption severity stress case
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Problem

Thailand's electronics, automotive, HDD/storage, and home-appliance industries depend on imported integrated circuits. This project measures how exposed these industries are to Taiwan-related semiconductor supply disruption using public trade data and transparent risk modeling.

Research Question

How exposed are Thai industries to Taiwan-related semiconductor supply chain disruption, and which industries face the highest potential impact?

Methodology

- TPDI:** measures Thailand HS8542 import dependency on Other Asia, nes, used as a Taiwan/Chinese Taipei proxy.
- Revenue-at-Risk:** converts trade dependency into industry-level semiconductor-linked exposure.
- Scenario Analysis:** applies 10%, 20%, and 40% disruption severity assumptions to base-case exposure.

Key Limitations

- Other Asia, nes is a proxy, not an official Taiwan-only bilateral figure.
- RAR and scenario impacts estimate exposure, not certain losses or forecasts.
- Industry chip-share assumptions are sensitivity inputs, not firm-level BOM data.

Tools and Outputs

Python, pandas, Plotly, Excel, Jupyter Notebook, UN Comtrade/WITS trade data. Outputs include cleaned datasets, notebooks, source code, interactive visualizations, final report, and recommendations.

Industry Exposure Summary

Industry	RAR Base	Severe	Risk Note
Electronics	5.77	2.31	Dominant aggregate exposure
Automotive	0.92	0.37	High bottleneck risk
HDD / Storage	0.87	0.35	Critical controller chips
Home Appliances	0.78	0.31	Moderate but rising intensity

Note: RAR Base and Severe values are in USD billions. Severe = 40% disruption severity applied to base-case Revenue-at-Risk.

Recommended Actions

- Build strategic inventory buffers for critical semiconductor components.
- Create a semiconductor supply-chain risk monitoring dashboard.
- Diversify suppliers where technically feasible, especially mature-node chips.
- Run annual semiconductor disruption scenario-planning exercises.
- Strengthen semiconductor-adjacent capabilities such as OSAT, advanced PCB, power electronics, and precision components.